

OSSP:

Commands:

uname : Displays basic operating system information.

lscpu : Displays CPU and processor details.

```
[vigneshkumar@vigneshs-MacBook-Air ~ % nano Week-2_T1.c
```

```
[vigneshkumar@vigneshs-MacBook-Air ~ % ./Week-2_T1  
Enter a Linux command: uname
```

```
Parent Process  
Parent PID : 3361  
Child PID : 3362
```

```
Child Process  
Child PID : 3362  
Parent PID: 3361
```

```
Executing command: uname
```

```
Darwin
```

```
Child process completed.  
vigneshkumar@vigneshs-MacBook-Air ~ % █
```

ps : Displays the currently running processes.

```
[vigneshkumar@vigneshs-MacBook-Air ~ % ./Week-2_T1  
Enter a Linux command: ps
```

```
Parent Process  
Parent PID : 3381  
Child PID : 3382
```

```
Child Process  
Child PID : 3382  
Parent PID: 3381
```

```
Executing command: ps
```

PID	TTY	TIME	CMD
3339	ttys000	0:00.03	-zsh
3381	ttys000	0:00.01	./Week-2_T1

```
Child process completed.  
vigneshkumar@vigneshs-MacBook-Air ~ % █
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system manages the CPU by dividing its processing time among multiple applications.

Example: When we open Chrome and a music player together, the OS schedules CPU time for both. It Process Both Chrome And music player

2. Memory (RAM) -> Memory Management

The operating system assigns a separate portion of RAM to every running application.

Example: Chrome and VS Code each get their own memory space while running.

3. Storage (SSD) → File System Management

The operating system stores and organizes files on the storage device.

Example: Saving a document in the Documents folder on our computer.

4. I/O Devices → Device Management

The operating system uses device drivers to control input and output devices.